

Problem 1

Let X and Y be metric spaces and let $f : X \rightarrow Y$ be a function. Assume that A and B are open subsets of X such that the restrictions $f|_A$ and $f|_B$ are continuous.

- (a) Show that $f|_{A \cup B}$ is continuous.
- (b) Is this result still true if A and B are both closed subsets of X ?
- (c) Is the result true for the union of an infinite number of open sets?
- (d) Is the result true for the union of an infinite number of closed sets?

Solution. (a): $V \subset Y$ open. $(f|_{A \cup B})^{-1}(V) := \{x \in A \cup B : f(x) \in V\} = (f|_A)^{-1}(V) \cup (f|_B)^{-1}(V)$ which is open because both parts are open (this comes from continuity of $f|_A$ and $f|_B$).

Problem 2 : Basic Exam, Fall 2017

Let $(a_n)_{n \geq 1}$ be a decreasing sequence of nonnegative numbers such that

$$\sum_{n=1}^{\infty} a_n < \infty.$$

Show that $na_n \rightarrow 0$ as $n \rightarrow \infty$.

Solution. Let $S_n := \sum_{i=1}^n a_i$ and by assumption $S_n \rightarrow L < \infty$. Since convergence implies Cauchy, $\forall \varepsilon > 0$ there exists $m \geq n \geq N_0$ where

$$S_m - S_n = \sum_{i=1}^{m-n} a_{n+i} < \frac{\varepsilon}{2} \Rightarrow (m-n)a_{n+(m-n)} < \frac{\varepsilon}{2}.$$

Let N_1 where $a_n < \frac{\varepsilon}{2N_0}$. For $M = \max(N_0 + 1, N_1)$ and $n \geq M$:

$$na_n = N_0 a_n + (n - N_0) a_n < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Problem 3

Let X be a metric space which is separable, i.e. X admits a countable dense subset. Show that

- (a) X is second countable, that is, there exists a countable base for the topology. Recall that a collection $\mathcal{B}$ of open sets is a base for the topology if each open set in X can be written as a union of elements in $\mathcal{B}$.
- (b) X has the Lindelöf covering property: given a collection of open sets $(G_\alpha)_{\alpha \in I}$ in X , there exists $I_1 \subseteq I$ countable such that

$$\bigcup_{\alpha \in I} G_\alpha = \bigcup_{\alpha \in I_1} G_\alpha.$$

Hint: Consider $\mathcal{B}_1 = \{V \in \mathcal{B}; V \subseteq G_\alpha \text{ for some } \alpha \in I\}$.

Solution. (a): Let $A \subset X$ be the countable dense subset of X . We want to show that

$$\mathcal{B} := \{B_q(a)\}_{(q,a) \in \mathbb{Q} \times A}$$

is a basis for X . $\forall U \in \mathcal{T}(X)$, $\forall x \in U$ there exists a $\delta > 0$ such that $B_\delta(x) \subseteq U$. By density $\exists a_x \in B_{\frac{\delta}{4}}(x)$. $\exists q_x \in \mathbb{Q} \cap (\frac{\delta}{4}, 3\frac{\delta}{4})$. Note $x \in B_{q_x}(a_x)$ and $\forall z \in B_{q_x}(a_x)$

$$d(x, z) \leq d(x, a_x) + d(a_x, z) < \frac{\delta}{4} + q_x < \frac{\delta}{4} + 3\frac{\delta}{4} \leq \delta$$

Thus $x \in B_{q_x}(a_x) \subset B_\delta(x) \subset U$.

$$U = \bigcup_{x \in U} B_{q_x}(a_x)$$

(b): You create $\mathcal{B}_1$ which is a open cover of X then you form I_1 by doing the following for all $V \in \mathcal{B}_1$ take one element out of $\{\alpha \in I : V \subset G_\alpha\}$. Since $\mathcal{B}_1$ is a open cover this set is also a open cover and is countable.

Problem 4

Let X be a separable metric space and let $A \subseteq X$ be uncountable. Show that there exists $a \in X$ such that for each open neighborhood V of a , the set $V \cap A$ is uncountable. (Such a point is called a *condensation point* of A .)

Solution. Assume the contrary there exists no condensation point for A in X . Thus $\forall x \in X$ any open set $V \in \mathcal{T}(X)$ has at most countable intersection with A . Pick a open set V_x for all x such that $x \in V_x$. Then $\{V_x\}_{x \in X}$ forms a open cover of x . There exists $X_1 \subset X$ where X_1 is countable by the Lindelöf covering and $\{V_x\}_{x \in X_1}$ is a open cover.

$$A = \bigcup_{x \in X_1} V_x \cap A$$

This is a countable union of countable elements hence A must be countable, which contradicts our assumptions.